

**CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL
COIMBATORE**

**ENTRANCE EXAM: SAMPLE PAPER
(FOR ADMISSION TO ACADEMIC YEAR 2014)**

CLASS: XI IB

SUBJECT: MATHS

01. Given positive integers 'a' and 'b', there exist unique integers 'q' and 'r' satisfying

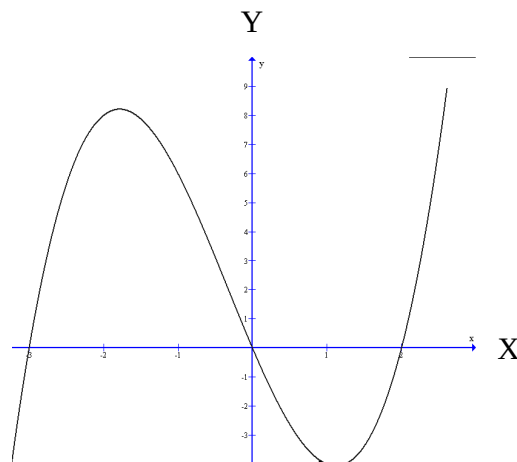
$a = bq + r$, where the values of 'r' are:

- A. $0 \leq r \leq a$ B. $0 \leq r \leq b$ C. $0 \leq r < b$ D. None of these

02. If the HCF of 96 and 404 is 4, then the LCM of 96 and 404 is

- A. 9636 B. 9696 C. 96404 D. None of these

03. The number of zeros in the following graph is



- A. 1 B. 2 C. 3 D. 4

04. If α and β are the zeros of the quadratic polynomial $p(x) = x^2 - 2$ then $\alpha + \beta$ is

- A. 0 B. 1 C. -2 D. None of these

05. The pair of lines; $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$ are

- A. intersecting B. coincident C. parallel D. None of these

06. Which term of the Arithmetic Progression: 21, 18, 15,is -81?

A. 35th

B. 15th

C. 25th

D. None of these

07. If $\sin 2A = 2 \sin A$ then $A =$ _____.

A. 30°

B. 0°

C. 45°

D. 60°